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PHYSICS

1)

Two particles are fired from the same point, with speeds 100 m/s and 100 m/s, and firing angles with horizontal = 60° and 120° respectively. The time after which their velocity vectors become perpendicular to each other, is :

(1) 5 s

(2) $5\left(\sqrt{3}-1\right)$ s

(3) $5\sqrt{3}s$

(4) $5\sqrt{3}/2s$

2) A man can swim in still water at a speed of 5 km/hr. He wants to cross a river 6 km wide, flowing at the rate of 4 km/hr. If he heads in a direction making an angle of 127° with stream direction, then he will reach a point on the other bank (approx) :-

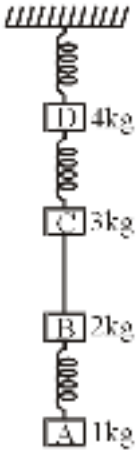
(1) upstream at a distance of 1.5 km

(2) downstream at a distance of 1.5 km

(3) directly on the other side of the bank

(4) never reach the other bank

3) In the given diagram, if the string connecting B and C is cut then what will be the accelerations of blocks A, B, C, D at that instant ?



(1) 0, g↓, g↑, 0

(2) $0, \frac{3g}{2}\downarrow, g\uparrow, 0$

(3) $0, \frac{3g}{2}\downarrow, \frac{3g}{2}\uparrow, 0$

(4) $g\downarrow, \frac{3g}{2}\uparrow, g\downarrow, \frac{5g}{4}\uparrow$

4) The engine of a motorcycle can produce a maximum acceleration 5 m/s^2. Its brakes can produce a maximum retardation 10 m/s^2.If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km :-
(Given : that motorcycle comes to rest at B)

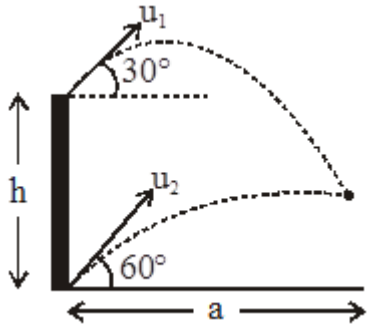
(1) 30 sec

(2) 15 sec

(3) 10 sec

(4) 5 sec

5) Shots are fired from the top of a tower and from its bottom simultaneously at angles 30° and 60° as shown. If horizontal distance of the point of collision is at a distance 'a' from the tower then height of tower h is :



(1) $\frac{2a}{\sqrt{3}}$

(2) $\frac{a}{\sqrt{3}}$

(3) 2a

(4) $\frac{4a}{\sqrt{3}}$

6) If $y = x - \frac{1}{2}x^2$ is equation of trajectory then time of flight is (g is acceleration of objects)

(1) $\sqrt{2g}$

(2) $\frac{2}{\sqrt{g}}$

(3) g

(4) $2.7\sqrt{g}$

7) A projectile has a time of flight T and range R. If time of flight is doubled keeping angle of projection same, then what will be the new range :-

(1) R

(2) 2R

(3) R/2

(4) 4R

8) The equation of motion of a projectile are $x = 36t$ and $2y = 96t - 9.8t^2$. Angle of projection is :-

(1) $\sin^{-1}\left(\frac{4}{5}\right)$

(2) $\sin^{-1}\left(\frac{3}{5}\right)$

(3) $\sin^{-1}\left(\frac{4}{3}\right)$

(4) $\sin^{-1}\left(\frac{3}{4}\right)$

9) The co-ordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time t is given by-

(1) $3t\sqrt{\alpha^2 + \beta^2}$

(2) $3t^2\sqrt{\alpha^2 + \beta^2}$

(3) $t^2\sqrt{\alpha^2 + \beta^2}$

(4) $\sqrt{\alpha^2 + \beta^2}$

10) A particle located at $x = 0$ at time $t = 0$, starts moving along the positive x-direction with a velocity 'v' that varies as $v = \alpha\sqrt{x}$. the displacement of the particle varies with time as

(1) t^2

(2) t

(3) $t^{1/2}$

(4) t^3

11) The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is-

(1) $v_0 + 2g + 3f$

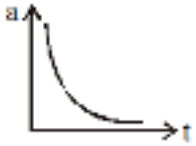
(2) $v_0 + g/2 + f/3$

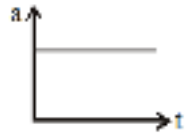
(3) $v_0 + g + f$

(4) $v_0 + g/2 + f$

12) The distance travelled by a body moving along a line in time t is proportional to t^3.
The acceleration-time (a, t) graph for the motion of the body will be :-

(1) 

(2) 

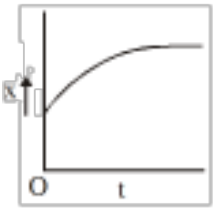
(3) 

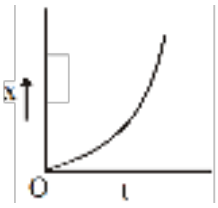
(4) 

13) A particle starts from origin O from rest and moves with a uniform acceleration along the positive x-axis. Identify all figures that correctly represent the motion qualitatively.
(a = acceleration, v = velocity, x = displacement, t = time)

(A) 

(B) 

(C) 

(D) 

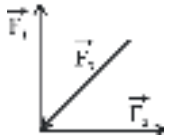
(1) (A), (B), (C)

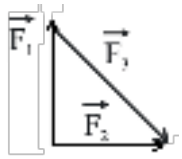
(2) (A)

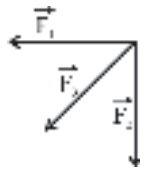
(3) (A), (B), (D)

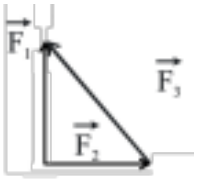
(4) (B), (C)

14) Which of the four arrangements in the figure correctly shows the vector addition of two forces $\vec{F_1}$ and $\vec{F_2}$ to yield the third force $\vec{F_3}$:-

(1) 

(2) 

(3) 

(4) 

15) A block can slide on a smooth inclined plane of inclination θ kept on the floor of a lift. When the lift is descending with a retardation a, the acceleration of the block relative to the incline is:-

(1) (g + a)sinθ

(2) (g – a)

(3) gsinθ

(4) (g – a)sinθ

16) Two iron blocks of equal mass but with double surface area slide down an inclined plane with friction coefficient μ. If the first block with surface area A experience a frictional force f, then the second block with surface area 2A will experience a frictional force :-

(1) f/2

(2) f

(3) 2f

(4) 4f

17) A player kicks a football of mass 0.5 kg and the football begins to move with a velocity of 10 m/s. If the contact between the leg and the football lasts for $\frac{1}{50}$ sec, then the force acted on the football should be :-

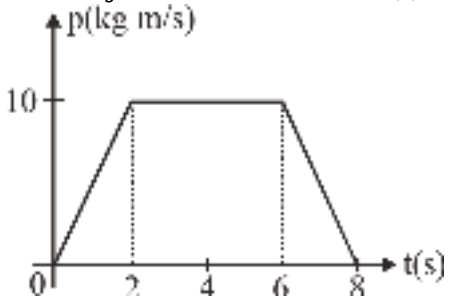
(1) 2500 N

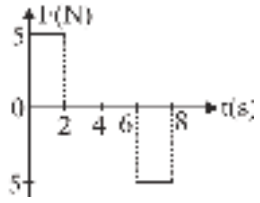
(2) 1250 N

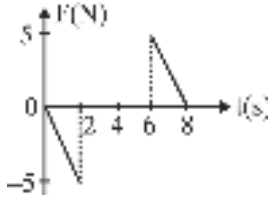
(3) 250 N

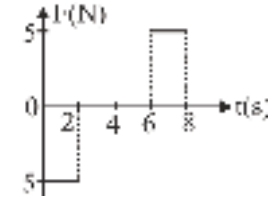
(4) 625 N

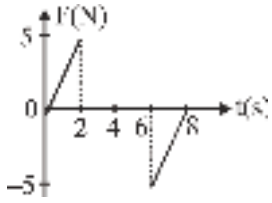
18) The momentum p of an object varies with time (t) as shown in figure. The corresponding force (F) time (t) graph is :-

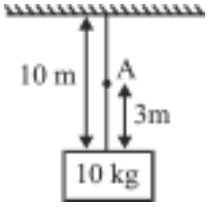


(1) 

(2) 

(3) 

(4) 



19) If mass of rope is 10 kg, the tension in the rope at point A is ($g = 10\ m/s^2$):-

(1) 170 N

(2) 30 N

(3) 130 N

(4) 70 N

20) Suppose a rocket with an initial mass M_0 expells at constant rate a mass Δm in the form of gases in time Δt, then the mass of the rocket after time t is :-

(1) $M_0 - \frac{\Delta m}{\Delta t} \cdot t$

(2) $M_0 - \frac{\Delta m}{\Delta t}$

(3) $M_0 + \frac{\Delta m}{\Delta t}$

(4) M_0

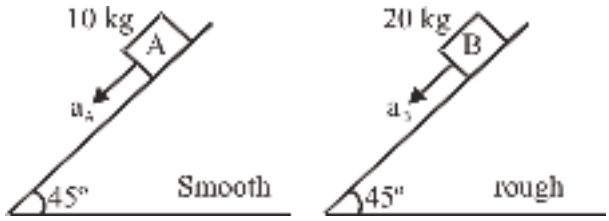
21) A body is thrown vertically upwards and takes 5 seconds to reach maximum height. The distance travelled by the body will be same in:-

(1) 1st and 10th second

(2) 2nd and 8th second

(3) 4st and 6th second

(4) Both (2) & (3)

22) 

The ratio of acceleration of block A placed on smooth incline with block B placed on rough incline is 2 : 1. The coefficient of kinetic friction between block B and incline is :-

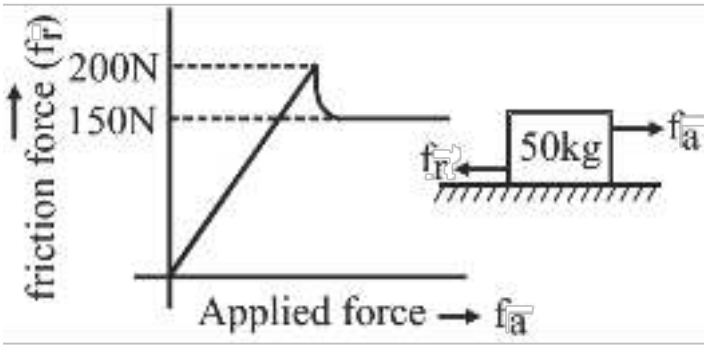
(1) 0.5

(2) 0.75

(3) 0.57

(4) None of these

23) The graph between applied force and frictional force for a body of mass 50 kg placed on a horizontal surface is as shown in figure. Then, what will be the value of coefficient of static and kinetic friction respectively ?

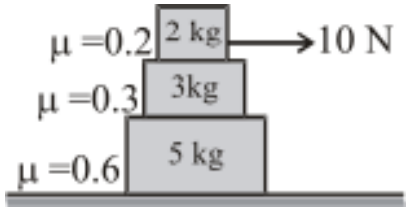


(1) 0.4, 0.3

(2) 0.3, 0.2

(3) 0.4, 0.2

(4) 0.2, 0.3

24) 

In the above figure acceleration of 2 kg block will be -

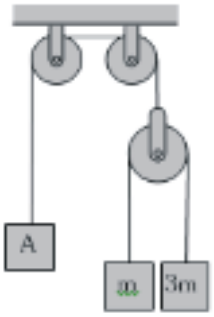
(1) 3 m/sec^2

(2) 2 m/sec^2

(3) 1 m/sec^2

(4) None

25) In the given figure, find mass of the block A, if it remains at rest, when the system is released



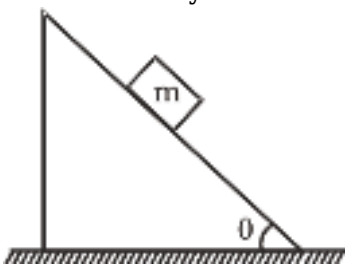
from rest. Pulleys and strings are ideal.

(1) m
(2) 2m
(3) 2.5m
(4) 3m

26) A spring-mass system vibrates such that mass travel on surface of coefficient of friction μ. Mass is released after compressing the spring by distance a and it travels upto distance b after its equilibrium position, then travelling from x = –a to x = b the reduction in its amplitude will be :-

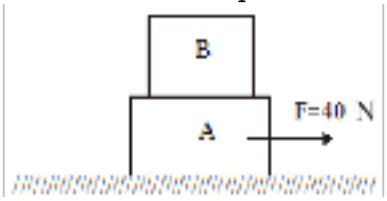
(1) $\frac{\mu mg}{K}$
(2) $\frac{2\mu mg}{K}$
(3) $\frac{\mu g}{K}$
(4) $\frac{K}{\mu mg}$

27) If block is sliding down on a rough fixed inclined plane with constant velocity as shown in the figure then net force acting on the inclined plane due to block is:-



(1) $\frac{mg}{\cos\theta}$
(2) mg
(3) mgcosθ
(4) None of these

28) A block B of mass 5 kg is placed on another block A of mass 10 kg, which rests on a smooth horizontal surface. If μ = 0.2 between A and B and a force F = 40N is applied, the acceleration of A is –



(1) 4 m/s^2

(2) 2 m/s^2

(3) 3 m/s^2

(4) 5 m/s^2

29) Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

(1) 9 min, 40 km/h

(2) 25 min, 100 km/h

(3) 10 min, 90 km/h

(4) 15 min, 120 km/h

30) The kinetic energy acquired by a body of mass m in travelling some distance s, starting from rest under the action of a constant force, is directly proportional to

(1) m°

(2) m

(3) m^2

(4) $\sqrt{m}$

31) A force $\vec{F} = -k(x\hat{i} + y\hat{j})$, where k is a constant, acts on a particle moving in the x-y plane. Starting from the origin, the particle is taken from the origin (0, 0) to the point (a, 0) and then from the point (a, 0) to the point (a, a). The total work done by the force on the particle is :

(1) ka^2

(2) $-ka^2$

(3) $2ka^2$

(4) $-2ka^2$

32) A block of mass 5 kg slides down a rough inclined surface. The angle of inclination is 45º. The coefficient of sliding friction is 0.20. When the block slides 10 m, the work done on the block by force of friction is $-50\sqrt{x}\,J$. Find the value of x :-

(1) 2

(2) 4

(3) 6

(4) 8

33) An engine pumps up 100 kg of water through a height of 10 m in 5 s. Given that the efficiency of the engine is 60%. If g = 10 ms^{-2}, the power of the engine is :

(1) 2 kW

(2) 3.3 kW

(3) 4 kW

(4) 6 kW

34) A body of mass 3 kg is under a force, which causes a displacement in it given by S = $\frac{t^3}{3}$ (in m). Find the work done by the force in first 2 seconds

(1) 18 J

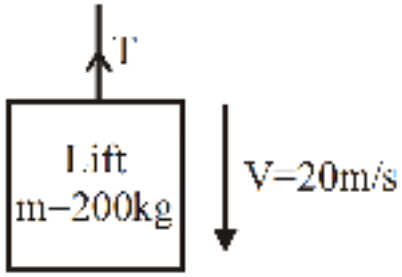
(2) 20 J

(3) 24 J

(4) 28 J

35) If this lift is moving downward with constant velocity 20 m/s, find work done by tension in 2 sec

:-



(1) -8×10^4 J

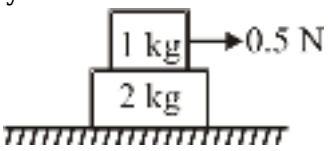
(2) -20×10^4 J

(3) -4×10^4 J

(4) -12×10^4 J

36) A force of 0.5 N is applied on the upper block as shown in figure. The coefficient of static friction between the two blocks is 0.1 and that between the lower block and the surface is zero. The work done by the lower block on the upper block for a displacement of 3 m of the upper block is –xJ. Then

x is : -



(1) 1 Joule

(2) –0.3 Joule

(3) –1 Joule

(4) +0.3 Joule

37) Two identical 5 kg blocks are moving with same speed of 2 m/s towards each other along a frictionless horizontal surface. The two blocks collide, stick together and come to rest. Consider the two blocks as a system. The work done by external and internal forces are respectively :

(1) 0, 0

(2) 0, 20 J

(3) 0, –20 J

(4) 20 J, – 20 J

38) A 60 kg man is standing in a lift. Column-I lists different motion conditions of the elevator and Column-II provides the corresponding normal force (i.e. the force exerted by the floor on the person). Match the enteties in column-I with the appropriate value in column-II.

	Column-I		**Column-II**
A.	Elevator moving at constant speed	(I)	Force on the floor by the person = 600 N
B.	Elevator accelerating upward at 3 m/s^2	(II)	Force on the floor by the person = 780 N
C.	Elevator accelerating downward at 3 m/s^2	(III)	Force on the floor by the person = 420 N

(1) A-I, B-II, C-III

(2) A-II, B-I, C-III

(3) A-III, B-I, C-II

(4) A-III, B-II, C-I

39) A body of mass m accelerates uniformly to a speed v_0 in time t_0. The work done on the body till anytime t is:-

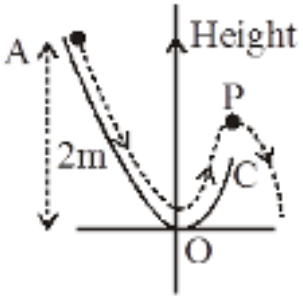
(1) $\frac{1}{2}mv_0^2\left(\frac{t^2}{t_0^2}\right)$

(2) $\frac{1}{2}mv_0^2\left(\frac{t_0}{t}\right)$

(3) $mv_0^2\left(\frac{t}{t_0}\right)$

(4) $mv_0^2\left(\frac{t}{t_0}\right)^3$

40) A particle (m = l kg) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaching its highest point P (height 1 m), the kinetic energy of the particle (in J) is : (Figure drawn is



schematic and not to scale; take g=10 ms^{-2})_____.

(1) 10 J

(2) 20 J

(3) 30 J

(4) 40 J

$v = k\sqrt{s}$

41) A locomotive of mass m starts moving so that its velocity varies according to the

law where k is constant and s is the distance covered. Find the total work performed by all the force which are acting on the locomotive during the first t second after the beginning of motion :-

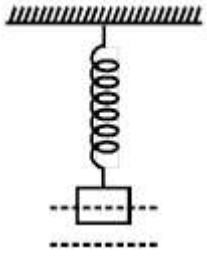
(1) $W = \frac{1}{8}mk^4t^2$

(2) $W = \frac{1}{4}m^2k^4t^2$

(3) $W = \frac{1}{4}mk^4t^4$

(4) $W = \frac{1}{8}mk^4t^4$

42) A block is attatched to a spring as shown and gradually lowered so that finally spring expands by "d". If same block is attached to spring & released suddenly then maximum expansion in spring will



be :-

(1) d

(2) 2d

(3) 3d

(4) 4d

43) Given $P = \frac{\alpha \cdot v}{(\beta + t)}$ when P is power, v velocity, t time dimensional formula of $\frac{\alpha}{\beta}$ is same as :-

(1) Force

(2) Work

(3) Momentum

(4) Pressure

44) The number of particles is given by $n = -D\left(\frac{n_2 - n_1}{x_2 - x_1}\right)$ crossing a unit area perpendicular to x-axis in unit time, where n_1 and n_2 are number of particles per unit volume for the value of x meant to x_1 and x_2. Find dimensions of D called as diffusion constant.

(1) $[M^0LT^2]$

(2) $[M^0L^2T^{-4}]$

(3) $[M^0LT^{-3}]$

(4) $[M^0L^2T^{-1}]$

45) 10 divisions on the main scale of a Vernier calliper coincide with 11 divisions on the Vernier scale. If each division on the main scale is of 5 units, the least count of the instrument is :

(1) $\frac{1}{2}$

(2) $\frac{10}{11}$

(3) $\frac{50}{11}$

(4) $\frac{5}{11}$

CHEMISTRY

1) Enthalpy of vaporisation for water is 186.5 kJ mol^{-1}. The entropy change during vaporisation of water is ..... kJ K^{-1} mol^{-1} :-

(1) 0.5

(2) 1.0

(3) 1.5

(4) 2.0

2) Enthalpies of formation of CO(g), CO_2(g), N_2O(g) and N_2O_4(g) are –110, –393, 81 and 9.7 kJ mol^{-1} respectively. Find the value of ΔH for the reaction :
$N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)$

(1) +778 kJ mol^{-1}

(2) –211.7 kJ mol^{-1}

(3) +211.7 kJ mol^{-1}

(4) –778 kJ mol^{-1}

3) Calculate the heat produced (in kJ) when 224 gm of CaO is completely converted to $CaCO_3$ by reaction with CO_2 at 27°C in a container of fixed volume.
Given : ΔH_f° ($CaCO_3$,s) = – 1207 kJ/mol ;
ΔH_f° (CaO, s) = – 635 kJ/mol ;
ΔH_f° (CO_2,g) = – 394 kJ/mol ;
[Use R = 8.3 J K^{-1} mol^{-1}]

(1) 702.04 kJ

(2) 821.96 kJ

(3) 1012 kJ

(4) 712 kJ

4) Heat of combustion of C_2H_6 and C_2H_2 is –341.1 and –310.0 Kcal mol^{-1}. Which one is best fuel.

(1) C_2H_6

(2) C_2H_2

(3) both

(4) None of these

5) Calculate P–Cl bond enthalpy
Given : $\Delta_f H(PCl_3,g)$ = 306 kJ/mol ;
$\Delta H_{atomization}$ (P,s) = 314 kJ/mol ;
$\Delta_f H(Cl,g)$ = 121 kJ/mol

(1) 123.66 kJ/mol
(2) 371 kJ/mol
(3) 19 kJ/mol
(4) 23 kJ/mol

6) Calculate the work done when 10 mol of ideal gas is compressed reversibly and isothermally from 10 L to 1 L at 27°C ?

(1) 57.44 kJ
(2) –57.44 kJ
(3) 1013 J
(4) –1013 J

7) For the given reaction :
$H_2(g) + S(s) \rightarrow H_2S(g)$; ΔH_r = 100 kJ/mol and ΔS_r = 400 J/mol/K
Temperature at which above reaction occurs reversibly is (Assumning ΔH_r and ΔS_r are independent of temperature)

(1) 200 K
(2) 250 K
(3) 400 K
(4) None

8) For the complete combustion of Methane,
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ in bomb calorimeter the amount of heat released is 150 kJ/mol at 127°C. Assuming ideality, the enthalpy of combustion would be :

(1) 150 kJ
(2) –150 kJ
(3) +147.5 kJ
(4) –152.5 kJ

9) The free energy change (ΔG) for a reaction having ΔH = 31400 cal, ΔS = 32 cal K^{-1} mol^{-1} at 1000°C is:-

(1) – 9336 cal
(2) – 7386 cal
(3) – 1936 cal
(4) + 9336 cal

10) When mixture of 6 L of CH_4 and Propane were burnt 7 L CO_2 was obtained then CH_4 and propane in mixture will be :-

(1) 5.5 L and 0.5 L

(2) 6 L and 1 L

(3) 3 L and 4 L

(4) 5 L and 2 L

11) In which case, a spontaneous reaction is impossible at any temperature :-

(1) ΔH = + ive, ΔS = + ive

(2) ΔH = + ive, ΔS = – ive

(3) ΔH = – ive, ΔS = – ive

(4) In all cases

12) What will be the concentration of acetate ion in a 1.0 M solution of acetic acid containing 0.1 M HCl [$K_a = 2 \times 10^{-5}$]:-

(1) 2×10^{-3} M

(2) 2×10^{-4} M

(3) 2×10^{-2} M

(4) 2×10^{-6} M

13) Assuming the reaction $A_2(g) + B_2(g) \rightleftharpoons C(g)$ to be exothermic, the yield of the product (C) increases with :-
(a) Increase of temperature
(b) Increase of pressure
(c) Addition of catalyst
(d) Addition of inert gas at constant volume
(e) Removal of C(g)

(1) a, b, c, d

(2) Only b

(3) b & e

(4) a & e

14) The equilibrium constant for reaction
$N_2 + O_2 \rightleftharpoons 2NO$ is K then the degree of reaction (α) will be :-

(1) $\sqrt{K}+1$

(2) $\sqrt{K}-1$

(3) $\frac{\sqrt{K}}{\sqrt{K}+2}$

(4) $\frac{1-\sqrt{K}}{\sqrt{K}}$

15) At 2000 K the equilibrium constant K_p, for the formation of NO(g)
$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4.0×10^{-4}
If at 2000K, the partial pressures of different species : $P_{N_2} = 0.5$ atm, $P_{O_2} = 0.2$ atm and $P_{NO} = 4\times10^{-3}$ atm :
Predict the system :-

(1) Will be at equilibrium
(2) Will go in the forward direction
(3) Will go in the backward direction
(4) None

16) **Assertion :-** pH of water increases with increase in temperature.
Reason :- Dissociation of water into H^+ and OH^- ion is an endothermic reaction.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

17) For the equilibrium $2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$,
calculate the ratio $\frac{K_P}{P}$, where P is the total pressure and $P_{Br_2} = \frac{P}{9}$ at a certain temperature at equilibrium.

(1) $\frac{1}{9}$
(2) $\frac{1}{81}$
(3) $\frac{1}{27}$
(4) $\frac{1}{3}$

18) For the following three reactions I, II and III, equilibrium constants are given :-
(I) $CO_{(g)} + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; K_1
(II) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$; K_2
(III) $CH_4(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 4H_2(g)$; K_3
Which of the following relations is correct ?

(1) $K_1\sqrt{K_2} = K_3$
(2) $K_2K_3 = K_1$
(3) $K_3 = K_1K_2$
(4) $K_3 = K_2^2 \cdot K_1$

19) Nicotinic acid ($K_a = 10^{-5}$) is represented by the formula HNiC. Calculate % dissociation in a solution which contains 0.1 mol of nicotinic acid per 2L of solution :-

(1) 1.4%

(2) 1.6%

(3) 2.4%

(4) 3.4%

20) 0.1M formic acid solution is titrated against 0.1M NaOH solution. What would be the difference in pH between 1/5 and 4/5 stages of neutralization of acid?

(1) 2log(3/4)

(2) 2log(1/5)

(3) log(1/3)

(4) 2log4

21) If 50 ml of 0.2 M KOH is added to 40 ml of 0.5M HCOOH. The pH of resulting solution is :- (K_a = 1.8×10^{-4})

(1) 3.75

(2) 5.6

(3) 7.5

(4) 3.4

22) The pH values of 0.1 M solution of HCOONa (I), HCOOH (II), NaCl(III), NaOH (IV), HCl (V) will be in the order :-

(1) IV > III > I > II > V

(2) IV > I > III > II > V

(3) II > III > I > IV > V

(4) V > II > III > I > IV

23) 0.2M NaOH is added to 50 mL of 0.2M acetic acid ($K_a = 1.8 \times 10^{-5}$). The volume of 0.2M NaOH required to make the pH of solution 4.74, is :-

(1) 11.5 mL

(2) 25 mL

(3) 37.5 mL

(4) 65 mL

24) K_{sp} of $Ca_3(PO_4)_2$ is :-

(1) $[Ca^{+2}][PO_4^{-3}]^2$

(2) $[Ca^{+2}]^3 [PO_4^{-3}]^2$

(3) $[Ca^{+2}]^2[PO_4^{-3}]^3$

(4) $[Ca^{+2}] [PO_4^{-3}]$

25) The degree of hydrolysis of salt of weak acid and weak base in its 0.1M solution is found to be 5%. If the molarity of the solution of same salt is 0.5 M, then the percentage degree of hydrolysis of

the salt should be :-

(1) 10 %
(2) 5 %
(3) 2.5%
(4) None of these

26) K_{sp} of $Mg(OH)_2$ is 1.8×10^{-11} then calculate solublity of $Mg(OH)_2$ in 0.1 M $Ba(OH)_2$ is :-

(1) 1.8×10^{-9}
(2) 0.9×10^{-11}
(3) 0.45×10^{-11}
(4) 0.45×10^{-9}

27) 10^{-4} M $AgNO_3$ and 10^{-6} M NaCl are mixed in 1:2 volume ratio, then precipitation of AgCl will [$K_{sp}(AgCl) = 10^{-6}$] :-

(1) Occur
(2) Not Occur
(3) 50% Occur
(4) 25% Occur

28)

Which of the following relation is incorrect

(1) $W_{\text{non expansion}} = -\Delta G$
(2) $W_{\text{useful}} = -\Delta G$
(3) $\Delta G_{\text{system}} = -T\Delta S_{\text{system}}$
(4) $K_{eq} = e^{-\frac{\Delta G^0}{RT}}$

29) The enthalpies of combustion of carbon and carbon monoxide are – 393.5 KJ and –283 KJ, respectively the enthalpy of formation of carbon monoxide is -

(1) – 676.5 KJ
(2) –110.5 KJ
(3) 110.5 KJ
(4) 676.5 KJ

30) **Assertion :** A spontaneous process is irreversible and may only be reversed by some external source.
Reason : Total entropy change (ΔS_{total}) of a spontaneous process is positive.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

31) Calculate $\Delta G°$ at 298 K for the reaction :
$2A_{(g)} + B_{(g)} \rightarrow 2D_{(g)}$
$\Delta U° = -10.5$ kJ and $\Delta S° = -44.1$ JK^{-1}

(1) 0.165 kJ
(2) 2.165 kJ
(3) 10.968 kJ
(4) 4.165 kJ

32) Latent heat of vaporization of a liquid is 20 kCal $mole^{-1}$ then calculate internal energy change for vaporization of 2 mole of a liquid at 227° C ?

(1) 20 kCal
(2) 40 kCal
(3) 60 kCal
(4) 38 kCal

33) 0.4g methane is combust in Bomb calorimeter at 25°C and 100 kJ heat is evolved in this process ΔE for the process is

(1) 4000 kJ/mol
(2) –4000 kJ/mol
(3) 3995 kJ/mol
(4) –3995 kJ/mol

34) Based on the B.E. value given :-
ΔH_f^o of $N_2H_{4(g)}$ is
Given N–N = 159 kJ/mol H–H = 436 kJ/mol
N≡N = 941 kJ/mol N–H = 398 kJ/mol

(1) 711 kJ $mole^{-1}$
(2) 62 kJ $mole^{-1}$
(3) –98 kJ $mole^{-1}$
(4) –711 kJ $mole^{-1}$

35) The heat of combustion of C_4H_{10} is –2878 kJ mol^{-1}. If the heats of formation of CO_2 and H_2O are –393.5 kJ mol^{-1} and –285.8 kJ mol^{-1} then the heat of formation of C_4H_{10} is :-

(1) –125.0 kJ mol^{-1}
(2) 250 kJ mol^{-1}
(3) –402.5 kJ mol^{-1}
(4) 402.5 kJ mol^{-1}

36) When 1L of CO_2 is heated with graphite, the volume of the gases collected is 1.5 L. Calculate the

number of moles of CO produced at S.T.P.:-

(1) $\frac{1}{11.2}$

(2) $\frac{28}{22.4}$

(3) $\frac{1}{22.4}$

(4) $\frac{14}{22.4}$

37) 10 g of a sample of $CaCO_3$ gave 3.3 g of CO_2 on treatment with an excess of HCl. The percentage purity of the sample is :-
$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

(1) 50

(2) 60

(3) 70

(4) 75

38) The number of mole of MnO_4^- required to completely oxidised 10 mole FeC_2O_4 in acidic medium..

(1) 2

(2) 4

(3) 5

(4) 6

39) When BrO_3^- ion reacts with Br^- in acidic medium, Br_2 is liberated. The equivalent weight of Br_2 in this reaction is :-

(1) $\frac{5M}{8}$

(2) $\frac{5M}{3}$

(3) $\frac{3M}{5}$

(4) $\frac{4M}{6}$

40) Which of the following reaction does not represent disproportionation reaction ?

(1) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

(2) $2H_2O_2 \rightarrow 2H_2O + O_2$

(3) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

(4) $2Cu^+ \rightarrow Cu^{2+} + Cu$

41) Match the column -

	Column-I (Reaction)		Column-II(Type of Reaction)
(A)	$2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$	(P)	Not Redox
(B)	$AgCl + KNO_3 \rightarrow AgNO_3 + KCl$	(Q)	Intermolecular Redox rection
(C)	$P_{4(s)} \xrightarrow{OH^-} PH_{3(g)} + H_2PO_2^-$	(R)	Disproportionation reaction
(D)	$H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$	(S)	Intramolecular Redox reaction

(1) A - R, B - P, C - R, D - S

(2) A - S, B - P, C - R, D - Q

(3) A - Q, B - P, C - R, D - S

(4) A - Q, B - R, C - P, D - S

42) Oxidation states of P in $H_4P_2O_5$, $H_4P_2O_6$, $H_4P_2O_7$ are respectively :

(1) +3, +4, +5

(2) +3, +5, +4

(3) +5, +3, +4

(4) +5, +4, +3

43) A gaseous mixture contains 0.80 mol SO_2Cl_2, 0.40 mol SO_2 and 0.20 mol Cl_2 in a 2 litre container. At 400 K, $K_c = 10$ for the reaction:-
$SO_{2(g)} + Cl_{2(g)} \rightleftharpoons SO_2Cl_{2(g)}$

(1) The system is at equilibrium

(2) More Cl_2, but no more SO_2, will form as the system achieves equilibrium

(3) More SO_2 and Cl_2 will form as the system achieves equilibrium.

(4) More SO_2Cl_2 will form as the system achieves equilibrium

44) Which of the following mixtures will act as buffer ?

(1) HCl + NaOH (1.5 : 3 molar ratio)

(2) HCN + NaOH (1.0 : 2 molar ratio)

(3) NH_4OH + HCl (4 : 1 molar ratio)

(4) NH_4OH + HCl (4 : 5 molar ratio)

45) Which indicator is suitable for NaOH and HCl titration :-

(1) Methyl orange

(2) Methyl red

(3) Phenolphthalin

(4) All

BIOLOGY

1) Which of the following phylum is devoid of tissue :-

(1) Porifera

(2) Coelentrata

(3) Annelida

(4) Ctenophora

2) Which of the following is true regarding molluscs :-

(1) Molluscs have soft bodies enclosed by chitinous shells

(2) Radula is present for feeding

(3) Adult molluscs are radially symmetrical

(4) Feather like gills are present near mouth.

3) How many structure from the following are haploid ?
(a) Prothallus of ferns
(b) Microsporophyll of *Selaginella*
(c) Rhizoids of prothallus
(d) Anthrezoids of ferns

(1) 1

(2) 2

(3) 3

(4) 4

4) Given below are two statements :
Statement I : In male frogs, ureters act as urinogenital ducts which open into cloaca.
Statement II : In Female froge, Ureters and oviducts open seperately into cloaca.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

5) Read the following statements carefully and choose the incorrect one.

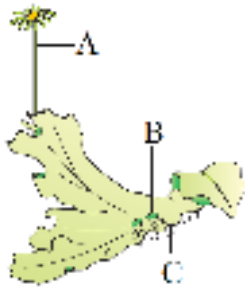
(1) Bacteria are the second largest members of Kingdom Monera

(2) Cyanobacteria are unicellular, colonial or filamentous forms

(3) Rod-shaped bacterium is bacilli

(4) BGA often forms blooms in polluted water bodies

6) Select the option that correctly identifies A, B and C in the give figure of female thallus of

Marchantia– 

(1) A – Gemma cup, B – Antheridiophore ,
C – Rhizoids

(2) A – Antheridiophore, B – Rhizoids,
C – Gemma cup

(3) A – Archegoniophore, B – Gemma cup,
C – Rhizoids

(4) A–Archegoniophore, B – Rhizoids,
C – Gemma cup

7) Which one of the following is an exclusive character of class-mammalia ?

(1) Homeothermy

(2) Internal fertilisation

(3) The presence of a four chambered heart

(4) The presence of a muscular diaphragm

8) The ureter act as urinogenital duct which opens into cloaca in–

(1) Only male frog

(2) Only female frog

(3) Both male and female frog

(4) Neither male nor female frog

9) **Statement-I :** Sexual spores are generally not found in members of Basidiomycetes but vegetative reproduction by fragmentation is common.
Statement-II : Members of deuteromycetes reproduce by asexual spore known as conidia.

(1) Only **statement-I** is correct

(2) Only **statement-II** is correct

(3) **Statements I & II** both are incorrect

(4) **Statements I & II** both are correct

10) Read the column-I, II and III and identify their correct match –

	Column-I		**Column-II**		**Column-III**
(A)	Chlorophyceae	(P)	Protonema	(i)	Pollination
(B)	Bryophyta	(Q)	Oogonium	(ii)	Mosses
(C)	Pteridophyta	(R)	Ovule	(iii)	Zygotic meiosis

(D)	Gymnosperm	(S)	Prothallus	(iv)	*Psilotum*

(1) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(2) (A)-(S)-(iv), (B)-(R)-(iii), (C)-(Q)-(ii), (D)-(P)-(i)

(3) (A)-(R)-(i), (B)-(S)-(iv), (C)-(P)-(ii), (D)-(Q)-(iii)

(4) (A)-(Q)-(iii), (B)-(P)-(ii), (C)-(S)-(iv), (D)-(R)-(i)

11) Select the **incorrect** statement regarding photosynthetic protista :-

(1) Diatoms are the chief producers in the oceans

(2) The spores are dispersed by air currents

(3) The pigments of euglenoids are identical to those present in higher plants

(4) Dinoflagellates have two flagella

12) **Statement-I :** Cartilage, bones and muscles are various types of specialised connective tissue .
Statement-II : Cartilage, bones and muscles are examples of connective tissue which provides strength, elasticity and flexibility of the body of complex animals.

(1) Statement-I is correct and statement-II is incorrect.

(2) Statement-I is incorrect and statement-II is correct.

(3) Both statement-I and statement-II are correct.

(4) Both statement-I and statement-II are incorrect.

13) Given below are two statements-
Statement-I : Cartilage a hard and non-pliable ground substance rich in calcium salts and collagen fibres.
Statement-II : Most of cartilages in vertebrate embryos are replaced by bones in adults.
In the light of above statements, choose the correct answer from the options given below.

(1) Statement-I is incorrect but statement-II is correct

(2) Both statement-I and statement-II are correct

(3) Both statement-I and statement-II are incorrect

(4) Statement-I is correct but statement-II is incorrect.

14) Find out incorrect statement :-

(1) Aschelminthes include parasitic as well as non parasitic forms.

(2) Agnatha is represented by the class cyclostomata.

(3) Sponges are primitive multicellular animals.

(4) Ctenidia helps in locomotion.

15) Read the following two statements (A and B) regarding members of monera and protista.
(A) Their cell wall composition is same
(B) Their cell type is different

(1) Statement A and B are correct

(2) Statement A is correct and B is incorrect

(3) Statement B is correct and A is incorrect

(4) Statement A and B are incorrect

16) Which of the following statement is not correct for mycoplasma ?

(1) Mycoplasmas are completely lack a cell wall.

(2) They are the smallest living cells.

(3) They can not survive without oxygen.

(4) Many mycoplasma are pathogenic in animals and plants.

17) Some gymnosperm have small specialised roots which are associated with N_2-fixing cyanobacteria. Name the gymnosperm and specialised roots.

(1) *Sequoia*, Tap root

(2) *Pinus*, Coralloid root

(3) *Cycas*, Coralloid root

(4) *Cedrus*, Adventituous root

18) Male reproductive system of frog consist of

(1) A pair of red coloured testes

(2) A pair of yellowish ovoid testes

(3) A pair of rounded testes

(4) Two pair of rounded testes

19) **Assertion (A) :** Archaebacteria are special since they live in some of the most harsh habitats such as extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens)
Reason (R) : Archaebacteria have a different complex cell wall and cell membrane structure than other bacteria.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(3) Both assertion and reason are incorrect.

(4) Assertion is correct but reason is incorrect.

20) Two basic forms of Coelenterates (*Obelia*) are given. Which option is correct ? (A)  (B) 

(1) (A) and (B) are free swimming

(2) (A) produce (B) asexually and (B) produce (A) sexually

(3) (A) is medusa and (B) is polyp

(4) (A) and (B) are sessile

21) Match the following organisms with their characteristics:-

(a)	*Psittaculla*	(i)	Notochord only in larval tail
(b)	*Calotes*	(ii)	Water vascular system
(c)	*Salpa*	(iii)	Air sacs connected to lungs
(d)	*Ophiura*	(iv)	Tympanum represent ear
(e)	*Sycon*	(v)	Water canal system

Select the **correct** option from the following:-

	(a)	**(b)**	**(c)**	**(d)**	**(e)**
(1)	(iv)	(ii)	(iii)	(i)	(v)
(2)	(iv)	(iii)	(ii)	(i)	(v)
(3)	(iii)	(iv)	(i)	(ii)	(v)
(4)	(iii)	(iv)	(i)	(v)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

22) Select the incorrect statement w.r.t. Frog–

(1) Protective colouration in frog is called mimicry

(2) Ventral side of skin is generally olive green

(3) Final digestion takes place in the intestine

(4) Only male possess sound producing prominent vocal sacs

23) Find out the incorrect statement–

(1) All the fungi are hetrotrophic

(2) Fungi prefer to grow in warm and humid places

(3) Wheat rust is cause by *Albugo*

(4) Yeast is used in making bread and beer

24) **Assertion-** In frog digestion of food take place by the action of HCl and gastric juices secreted from the wall of stomach.
Reason - Bile emulsifies fat and pancreatic juices digest carbohydrates and proteins.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are correct and Reason is not the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect

(4) Assertion is incorrect but Reason is correct

25) Which of the following mouth parts in cockroach are paired :

(1) Mandible and Labium

(2) Labium and Labrum

(3) Maxilla and Hypopharynx

(4) Mandible and Maxilla

26) A student was studying a plant and found the following features :
Main plant body is sporophyte
Root stem and leaf present
Archegonia present
Antheridium is present
Seed formation is absent
The plant belongs to the division :

(1) Bryophyta

(2) Pteridophyta

(3) Gymnospermae

(4) Angiospermae

27) **Statement-I :** In frog each kidney is composed of several structural and functional units called uriniferous tubules.
Statement-II : In frog a triangular structure called sinus venosus joins the left atrium.

(1) Both statement-I and statement-II are incorrect.

(2) statement-I is correct but statement-II is incorrect

(3) statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are correct

28) Bryophytes are also called 'amphibians of the plant kingdom' because

(1) water is essential for reproduction

(2) they occur only in water

(3) these plants can live in soil, but are dependent on water for sexual reproduction

(4) water is essential for spore formation

29) Match the column I, column II and column III and select the correct option from options given below :

	Column I (Types of spores)		**Column II (Fungus)**		**Column III (Fungi class)**
(A)	Oopspores	(P)	*Penicillium*	(i)	Basidiomycetes
(B)	Conidia	(Q)	*Albugo*	(ii)	Deuteromycetes
(C)	Ascospore	(R)	*Agaricus*	(iii)	Ascomycetes
(D)	Basidiospore	(S)	*Trichederma*	(iv)	Phycomycetes

(1) A-S-ii, B-P-iii, C-Q-iv, D-R-i

(2) A-Q-iv, B-P-iii, C-S-ii, D-R-i

(3) A-Q-iv, B-S-ii, C-P-i, D-R-iii

(4) A-Q-iv, B-S-ii, C-P-iii, D-R-i

30) Which of the following statement is correct?

(1) Artificial classification system gave more weightage to sexual characteristics.

(2) The natural classification system gave equal weightage to vegetative and sexual characteristics.

(3) Phylogenetic classification system is based on evolutionary relationship between the various organisms.

(4) Artificial classification for flowering plants was given by George Bentham and Joesph Dolton Hooker.

31) Select the correct statements out of following :-
(A) Fungi association with roots of higher plants is called Mycorrhiza
(B) In phycomycetes mycelium is aseptate and multinucleate
(C) Ascomycetes are coprophilous
(D) The basidiospores are endogenously produced in basidium.
(E) *Puccinia* and *Colletotrichum* are example of deuteromyetes.
Options :-

(1) (A), (B) and (C)

(2) (C), (D) and (E)

(3) (A), (D) and (E)

(4) (B), (C) and (E)

32) Choose the correct match :

(1) Hemidactylus – Wall lizard

(2) Chameleon – Gardan lizard

(3) Calotes – Tree lizard

(4) Chelone – Tortoise

33) Given below are two statement :- One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : In nearly all animal tissue, specialised junctions provide both structural and functional links between individual cells.
Reason (R) : Gap junctions help to stop substances from leaking across a tissue.
In the light of the above statements, choose the correct answer from the option given below:-

(1) (A) is not correct but (R) is correct

(2) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(4) (A) is correct but (R) is not correct.

34) Read the following terms carefully :

Cilia, Flagella, Heterocyst, Unicellular, Eukaryotes, Photosynthesis, Embryo, Zygote, Tissue, Seed, Murein, Pili, Pellicle, Chemosynthesis, Nitrogen fixation.
How many terms are only and only related with Monera and not found in any other kingdom?

(1) 8
(2) 7
(3) 5
(4) 10

35) A plant has vascular tissue, forms cone and has archegonia and antheridia both. This plant can be :

(1) A gymnosperm
(2) A pteridophyta
(3) Either gymnosperm or pteridophyta
(4) Bryophyta or pteridophyta

36) Which of the following is heterosporous fern ?

(1) *Salvia*
(2) *Salvinia*
(3) *Selaginella*
(4) 2 and 3 both

37) **Assertion :** Water vascular system helps in locomotion, capture and transport of food and respiration in echinodermata.
Reason : Water vascular system is the most distinctive feature of phylum-enhinodermata.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

38) Match List-I with List-II

List-I		List-II	
(A)	Bryophyta	(I)	Gametophyte and sporophyte are independent
(B)	Pteridophyta	(II)	Reduced Gametophyte is dependent on sporophyte
(C)	Gymnosperms	(III)	Sporophyte is dependent on gametophyte
(D)	Angiosperms	(IV)	Extremely reduced Gametophyte is dependent on sporophyte

Choose the correct answer from the options given below :
(1) A-(III), B-(I), C-(II), D-(IV)

(2) A-(IV), B-(III), C-(II), D-(I)

(3) A-(I), B-(II), C-(III), D-(IV)

(4) A-(II), B-(IV), C-(I), D-(III)

39) Which bacteria oxidise various inorganic substances such as nitrites and ammonia and use the released energy for their ATP production ?

(1) Photosynthetic autotrophic

(2) Heterotrophic

(3) Chemosynthetic autotrophic

(4) Mycoplasma

40) Feature related to frog is

(1) Poisoning gland in mouth

(2) Show Hibernation

(3) Placoid scales on body

(4) Neurogenic heart

41) A. Unique kingdom of heterotrophic organisms.
B. Show a great diversity in morphology and habitat.
C. Prefer to grow in warm and humid places.
D. Cosmopolitan and occur in air, water, soil.
Above Statements are about Which kingdom ?

(1) Kingdom Protista

(2) Kingdom Plantae

(3) Kingdom Fungi

(4) Kingdom Animalia

42) Select the incorrect statement with respect to bryophytes :-

(1) Thallus is attached to the substratum by rhizoids.

(2) The sporophyte is free-living

(3) Sex organs are multicellular

(4) Zygotes do not undergo reduction division immediately.

43) Mesorchium is a double fold of peritoneum which connect :-

(1) Testes to kidney

(2) Ovary to kidney

(3) Testes to liver

(4) Ovary to liver

44) **Statement I :** Fimbriae are small bristles-like fibres to help the bacteria to attach on rocks and to host tissues.
Statement II : Chromatophores are membraneous extensions with photosynthetic pigments present

in some bacteria.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

45) **Statement-I :** Gemma are green, multicellular asexual buds, which develop in small receptacles called gemma cups located on the thalli.
Statement-II : The predominant stage of the life cycle of a liverworts is the gametophyte which consists of two stages. The first stage is protonema stage & the second stage is the leafy stage.

(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statement I & II is correct
(4) Both statement I & II is incorrect

46) Ducts of glands and tubular part of nephrons in kidney having which type of epithelium ?

(1) Simple squamous
(2) Simple cuboidal
(3) Simple coloumnar
(4) Pseudostratified epithelium

47) Which one is the wrong statement about the "bones" ?

(1) The bone-marrow in some bones is the site of production of blood-cells.
(2) Bones have solid and pliable ground substance.
(3) Matrix of bones is rich in Ca^{+2} salts and collagen fibres.
(4) Bone cells are present in the spaces called lacunae.

48) Select the correct option for statement-I & statement-II.
Statement-I : Worm like marine cylindrical animals included in Phylum Hemichordata.
Statement-II : In echinodermates mouth present on lower side and anus present on upper side.

(1) Statement-I is correct, statement-II is incorrect.
(2) Statement-I is incorrect, statement-II is correct.
(3) Statement-I & II both are incorrect.
(4) Statement-I & II both are correct.

49) **Assertion (A) :** The leaves of gymnosperms are well adapted to withstand extremes of temperature, humidity and wind.
Reason (R) : In gymnosperms the needle like leaves reduce surface area. Their thick cuticle and sunken stomata also help to reduce water loss.

(1) Both A and R are correct but R is not the correct explanation of A.
(2) A is correct but R is not correct.

(3) A is not correct but R is correct.
(4) Both A and R are correct and R is correct explanation of A.

50) The correct pathway of blood circulation in cockroach :-

(1) Sinus → Ostia → Heart → Anterior to body
(2) Sinus → Ostia → Heart → Posterior to body
(3) Posterior → Heart → Ostia → Sinus
(4) Heart → Ostia → Sinus → Anterior to body

51) Find monoecious animals in the given list :-
Nereis, *Ancylostoma*, Leeches, *Laccifer*, *Pinctada*

(1) Two
(2) Three
(3) One
(4) Five

52) "Animals exhibits radial or bilateral symmetry depending on the stage".
This statement is true for :-

(1) Echinodermates
(2) Molluscans
(3) Arthropods
(4) Hemichordates

53) Zoospores are formed in a particular class of fungi in which the members :

(1) Have septate and uninucleate mycelia
(2) Produce exogenous spores only
(3) Do not show sexual reproduction
(4) Have coenocytic mycelium

54) Prothallus in pteridophyte is :-

(1) Vascular, Haploid, Multicellular sporophyte
(2) Diploid, Free living gametophyte
(3) Non vascular, haploid multicellular gametophyte
(4) Green multicellular, sporophyte

55) RBC’s of Frog contain–

(1) Nucleus but not haemoglobin
(2) Haemoglobin but not nucleus
(3) Both nucleus and haemoglobin
(4) Plasma proteins

56) What is the right sequence of structure of mollusca from outside to inside -

(1) Shell → Mantle cavity → Mantle → Visceral hump.
(2) Mantle → Shell → Mantle cavity → Visceral hump.
(3) Shell → Mantle → Mantle cavity → Visceral hump.
(4) Shell → Mantle → Visceral hump. → Mantle cavity

57) Hairs are absent in which animal :

(1) Macropus
(2) Camelus
(3) Neophron
(4) Elephas

58) Select the correct route for the passage of sperms in male frog in reverse order (outside to inside)

(1) Cloaca → Urinogenital duct → Seminal vesicle → Kidney → Vasa efferentia → Testes
(2) Cloaca → Ureter → Bidder's canal → Vasa efferentia → Testes
(3) Cloaca → Urinogenital duct → Bidder's canal → Kidney → Vasa efferentia → Testes
(4) Cloaca → Urinogenital duct → Vasa efferentia → Kidney → Bidder's canal → Testes

59) All animals of which phylum are exclusively marine?

(1) Ctenophora
(2) Echinodermata
(3) Mollusca
(4) Both 1 and 2

60) Creeping, green, branched and frequently filamentous stage represents :

(1) Sporophyte of liverwort
(2) Gametophyte of liverwort
(3) First stage of gametophyte in mosses
(4) Second stage of gametophyte in mosses

61) Read the following statements carefully.
(A) In gymnosperms development of male gametophyte takes place within the microsporangia.
(B) Gymnosperms are heterosporous plants in which spores show germination in sporangia.
(C) In pteridophytes well distinct alternation of generation is present.
(D) In pteridophytes both gametophyte and sporophyte generation are independent.
(E) Bryophyte are of little ecological importance.
How many above statements are correct.

(1) One
(2) Two
(3) Three

(4) Four

62) In cockroach first pair of wings arises from ____(A)____ and the second pair from _____(B)_____. Identify A and B from given option :

(1) A - Prothorax, B - Mesothorax

(2) A - Prothorax, B - Metathorax

(3) A - Mesothorax, B - Prothorax

(4) A - Mesothorax, B - Metathorax

63) Match the column A with column B and find out the correct option as shown ?

	Column-A		**Column-B**
a.	Pollution indicator	p.	Fungal component
b.	Mycobiont	q.	Viroid
c.	Potato spindle tuber disease	r.	Lichen
d.	Obligate parasite	s.	Virus

(1) a-r, b-p, c-s, d-q

(2) a-r, b-s, c-p, d-q

(3) a-s, b-r, c-p, d-q

(4) a-r, b-p, c-q, d-s

64) **Assertion (A) :** The Malpighian tubules of the cockroach are enteronephric in nature.
Reason (R) : Malpighian tubules of cockroach absorb nitrogenous waste products from haemolymph and release them into the hind gut.
Choose the correct answer from the options given below :-

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

65) **Statement - I :** Cockroach eyes consist of several ommatidia. So they receive several images of an object.
Statement - II : Mosaic vision is present in cockroach with more sensitivity but less resolution.

(1) Both statement I and statement - II are incorrect.

(2) Statement - I is correct but statement - II is incorrect.

(3) Statement - I is incorrect but statement - II is correct.

(4) Both statement - I and statement - II are correct.

66) How many of following are incorrect regarding deuteromycetes.
I. Reproduce by sexual spore conidia.
II. Mycelium is branched and septate
III. All members are parasite only.

IV. Helps in mineral cycling

(1) I, II & III
(2) I, III
(3) II, III & IV
(4) II & IV

67) Read column-I, II & III and identify their correct match :-

Column-I		**Column-II**		**Column-III**	
(A)	Metameric segmentation	(P)	Porifera	(i)	*Sycon*
(B)	Spongin fibers	(Q)	Annelida	(ii)	*Pheretima*
(C)	Metagenesis	(R)	Mollusca	(iii)	*obelia*
(D)	Radula	(S)	Coelenterata	(iv)	*Pila*

(1) (A)-(Q)-(ii), (B)-(P)-(i), (C)-(S)-(iv), (D)-(R)-(iii)
(2) (A)-(R)-(ii), (B)-(P)-(i), (C)-(S)-(iii), (D)-(Q)-(iv)
(3) (A)-(R)-(iv), (B)-(Q)-(i), (C)-(S)-(ii), (D)-(P)-(iii)
(4) (A)-(Q)-(ii), (B)-(P)-(i), (C)-(S)-(iii), (D)-(R)-(iv)

68) Arrange the following different taxa in ascending sequence according to their rank.
A. Poaceae B. Polymoniales
C. *Panthera* D. Monocotyledonae

(1) C – A – B – D
(2) D – C – B – A
(3) C – B – A – D
(4) D – B – A – C

69) Read the following statements carefully :-
(1) In *Euglena*, protein rich layer called pellicle makes their body flexible.
(2) Mixotrophic mode of nutrition is found in Euglena.
(3) In Slime mould spores possess true walls.
(4) *Gonyaulax* are red dinoflagellates.
How many above statements are correct :-

(1) One
(2) Two
(3) Three
(4) Four

70) Which of the following group of marine algae is used as food.

(1) *Chlamydomonas*, *Volvox* and *Gracilaria*
(2) *Porphyra*, *Laminaria* and *Sargassum*

(3) *Laminaria* and *Gracilaria*

(4) *Porphyra* and *Chlamydomonas*

71) **Statement-I :** On an average female cockroach produce 9-10 ootheca.
Statement-II : The nymph grows by moulting about 13 times to reach the adult form.

(1) Statement-I and statement-II both are correct.

(2) Statement-I is correct and statement-II is incorrect.

(3) Statement-I is incorrect and statement-II is correct.

(4) Statement-I and statement-II both are incorrect.

72) Match the column I and Column II :

	Column-I		**Column-II**
(P)	*Gonyaulax*	(i)	Soap box like
(Q)	Slime moulds	(ii)	Red tides
(R)	Chrysophytes	(iii)	Released toxins for marine animals
		(iv)	Spread over several feet
		(v)	Spores survive for many years
		(vi)	Golden algae
		(vii)	Diatomaceous earth

(1) P → vi, vii; Q → iv, v ; R → i, ii, iii

(2) P → iv, v; Q → i, ii, iii ; R → vi, vii

(3) P → ii, iii; Q → iv, v ; R → i, vi, vii

(4) P → iv, v ; Q → vi, vii ; R → i, ii, iii

73) Some features of animals are given here
Organ system level, coelomate, open circulation, triploblastic, unsegmented, Intracellular digestion, Radial symmetry.
How many characters are present in the members of phylum Annelida.

(1) Two

(2) Three

(3) Four

(4) Five

74)

How many organisms in the list given below belong to ascomycetes ?
Aspergillus, Rhizopus, Yeast*, Claviceps,* Morels*, Agaricus, Trichoderma, Puccinia*

(1) Six

(2) Three

(3) Five

(4) Four

75) **Assertion :** Amphibians are unsuccessful terrestrial chordates.
Reason : Amphibian depends on water for their reproducton.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

76) Read the following statements :-
(a) Members of phylum porifera are known as sea walnuts or comb jellies.
(b) Flatworms are bilaterally symmetrical, triploblastic and coelomate animals.
(c) *Planaria* possess high regeneration capacity.
(d) Aquatic annelids like Nereis possess lateral appendages parapodia.
(e) Most distinctive feature of echinoderms is the presence of water transport system.
Find the correct statements.

(1) a, b, c and e only
(2) a, b, d and e only
(3) c and d only
(4) c, d and e only

77) How many plants are heterosporous from the following ?
Chara, Sphagnum, Spirogyra, Marchantia, Selaginella, Funaria, Salvinia, Cycas, Mango, *Pinus*

(1) 6
(2) 5
(3) 4
(4) 7

78) How many of the given below diseases are caused by virus ?
Cholera, Influenza, Typhoid, Mumps, Tetanus, Herpes, AIDS

(1) Two
(2) Three
(3) Four
(4) One

79) Which one is not correctly matched ?

(1) *Cedrus* - Unbranched stem
(2) *Pinus* - Mycorrhizae
(3) *Cycas* - Unbranched stem
(4) *Sequoia* - Red wood tree

80) Read column-I, II and III and identify their correct match.

Column-I		Column-II		Column-III	
(A)	Porifera	(P)	Flame cells	(i)	Defense
(B)	Coelenterata	(Q)	Spicules or spongin fibers	(ii)	Excretion and osmoregulation
(C)	Platyhelminthes	(R)	Cnidoblast	(iii)	Swimming
(D)	Annelida	(S)	Parapodia	(iv)	Form skelton

(1) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(2) (A)-(Q)-(iv), (B)-(R)-(i), (C)-(P)-(ii), (D)-(S)-(iii)

(3) (A)-(R)-(ii), (B)-(S)-(iv), (C)-(Q)-(i), (D)-(P)-(iii)

(4) (A)-(S)-(iii), (B)-(P)-(ii), (C)-(R)-(iv), (D)-(Q)-(i)

81) Select the correct statements from the following :-
(A) Sexual reproduction in green algae is by flagellated zoospores.
(B) The predominant stage of the life cycle of a moss is the gametophyte.
(C) The main plant body of *Selaginella* is differentiated into true root, stem and leaves.
Options :-

(1) Only (A)

(2) Both (B) & (C)

(3) Only (B)

(4) Both (A) & (C)

82) **Statement-I :** *Gonyaulax* spreads on surface of sea water through rapid multiplication due to which the sea water appear red
Statement-II : Dinoflagellate have two flagella one is transverse and other is longitudinal

(1) Both statement I and II correct

(2) Both statement I and II incorrect

(3) Statement I is correct and II incorrect

(4) Statement I is incorrect and II correct

83) Match list-I with list-II :

	List-I		List-II
(A)	Phaeophyceae	(I)	Non-motile spores and gametes
(B)	Rhodophyceae	(II)	2-8 equal flagella in spores and gametes
(C)	Chlorophyceae	(III)	Single egg in flask shaped multicellular archaegonium
(D)	Bryophyta	(IV)	Unequal biflagellated pear shaped male gametes

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-I, C-II, D-III

84) In __A__ the male and female cones or strobili are borne on the same tree while in __B__ male cones and megasporophylls are borne on different tress. A & B are respectively :

(1) *Cycas, Pinus*

(2) *Pinus, Cycas*

(3) *Cycas, Ginkgo*

(4) *Ginkgo, Pinus*

85) What is the function of cilia :-

(1) Absorption

(2) To connect the structures

(3) Secretion

(4) To move particles and mucus in a specific direction over epithelium

86) Which of the following animals is a true fish

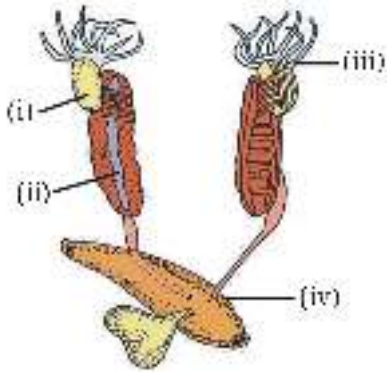
(1) Star fish

(2) Sea horse

(3) Prawn

(4) Cuttle fish

87) Given below is the diagram of reproductive system of an animal labelled by (i), (ii), (iii) and (iv),



identify the animal with correct labelling of (i) to (iv)

(1) Male frog-(i) Testes (ii) Adrenal gland (iii) Fat bodies (iv) Cloaca

(2) Female frog-(i) Kidney (ii) Adrenal gland (iii) Ovary (iv) Cloaca

(3) Male frog-(i) Testes (ii) Bidder canal (iii) Adrenal gland (iv) Cloaca

(4) Female frog-(i) Ovary (ii) Oviduct (iii) Adrenal gland (iv) Cloaca

88) Excretory structures in cockroach is/are :

(1) Malpighian tubules

(2) Fat body

(3) Nephrocyte, Urecose gland

(4) All of these

89) Match the following characteristic with respective animal :

	Column-I		**Column-II**
(A)	Radula	(I)	Planaria
(B)	Muscular pharynx	(II)	Limulus
(C)	Abscence of Anus	(III)	Pila
(D)	Jointed appendages	(IV)	Wuchereria

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-II, B-III, C-I, D-IV

90) Which one of the following statements is right ?

(1) Agar-Agar is obtained from brown algae.

(2) *Neurospora* used as space food.

(3) Mannitol is stored food in red algae.

(4) Algin and carragen are product of algae.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	2	1	1	2	4	1	2	1	2	4	3	3	1	2	3	1	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	1	1	4	2	2	3	4	1	2	1	2	3	1	3	3	1	1	1
Q.	41	42	43	44	45															
A.	1	2	1	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	1	2	1	1	2	2	1	1	2	2	3	3	2	3	2	3	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	2	2	2	4	2	3	2	2	1	4	2	2	1	3	4	4	3	1
Q.	86	87	88	89	90															
A.	2	1	3	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	2	3	4	1	3	4	1	2	4	2	4	1	4	3	3	3	2	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	3	2	4	2	2	3	4	3	1	1	4	3	2	2	2	1	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	1	4	1	2	2	4	4	1	3	1	4	3	3	3	3	3	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	4	2	4	2	4	1	4	2	1	3	2	4	1	3	2	3	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	1	4	2	4	2	1	4	2	4										

SOLUTIONS

PHYSICS

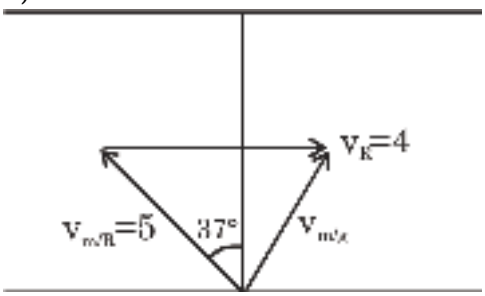
1) $\left[50\hat{i}+\left(50\sqrt{3}-10t\right)\hat{j}\right]\cdot\left[-50\hat{i}+\left(50\sqrt{3}-10t\right)\hat{j}\right]=0$

$-2500+\left(50\sqrt{3}-10t\right)^2=0$

on solving

$t=5\left(\sqrt{3}-1\right)$

2) $\bar{v}_{m/g}=\hat{i}+4\hat{j}$



$v_y = 4$

$v_x = 1$

$t = \frac{6}{4} = \frac{3}{2}hr$

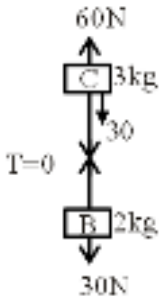
$D = v_x \times t = \frac{3}{2}km$

3) $a_B = \frac{30}{2} = 15 = \frac{3g}{2}\downarrow$

$a_C = \frac{60-30}{3} = 10 = g\uparrow$

$a_A = 0$

$a_D = 0$



4) $S = \frac{1}{2}\left(\frac{\alpha\beta}{\alpha+\beta}\right)t^2$

t = 30 s

5) $a = u_1\cos 30\, t = u_2 \cos 60\, t$
& $y = u_2 \sin 60\, t - (1/2)\, gt^2$
& $-(h - y) = u_1 \sin 30\, t - (1/2)gt^2$
$\Rightarrow y = a \tan 60 - (1/2)gt^2$
$\Rightarrow\ y - h = a \tan 30 - (1/2)gt^2$
$\therefore\ h = a(\tan 60 - \tan 30) \quad \Rightarrow h = 2a/\sqrt{3}$

6) $y = x\tan\theta - \frac{1}{2}g\frac{x^2}{u^2\cos^2\theta}$
$y = x \cdot 1 - \frac{1}{2} \cdot \frac{x^2}{1}$
$\therefore\ \tan\theta = 1 \quad \theta = 45°$
$\frac{u^2\cos^2\theta}{g} = 1$
$u = \frac{\sqrt{g}}{\cos\theta} = \sqrt{2g}$
$T = \frac{2u\sin\theta}{g} = \frac{2\sqrt{2g}\frac{1}{\sqrt{2}}}{g} = \frac{2}{\sqrt{g}}$

7) $T \propto u \quad R \propto u^2$

8) $x = u\cos\theta t \qquad y = u\sin\theta t - \frac{1}{2}gt^2$
$x = 36t \qquad y = \frac{96}{2}t - \frac{1}{2}(9.8)t^2$
$u\cos\theta = 36$
$u\sin\theta = \frac{96}{2}$
$\therefore\ \tan\theta = \frac{96/2}{36} = \frac{4}{3}$
$\therefore\ \sin\theta = \frac{4}{5}$

9)

$x = \alpha t^3 \quad y = \beta t^3$
$\vec{r} = x\hat{i} + y\hat{j}$
$\vec{v} = \frac{d\vec{r}}{dt} = 3\alpha t^2\hat{i} + 3\beta t^2\hat{j}$
$\text{speed} = 3t^2\sqrt{\alpha^2 + \beta^2}$

10) $v = \alpha\sqrt{x}$
$\frac{dx}{\sqrt{x}} = \alpha\, dt$

$$\frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \alpha t$$
$$2\sqrt{x} = \alpha t$$
$x \propto t^2$
$v \propto t$

11)

$v = v_0 + gt + ft^2$
$$\frac{dx}{dt} = v_0 + gt + ft^2$$
$$\int_0^x dx = \left(v_0 t + \frac{gt^2}{2} + \frac{ft^3}{3}\right)_0^1$$
$$\boxed{x = v_0 + \frac{g}{2} + \frac{f}{3}}$$

12)

$x \propto t^3$
$v \propto t^2$
$a \propto t$

13)

$a \rightarrow$ constant
$v \propto t^1$
$x \propto t^2$

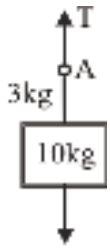
14) $\vec{F}_1 + \vec{F}_2 = \vec{F}_3$ (Triangle law)

15) Acceleration of block in a stationary lift = $g\sin\theta$
If lift is descending with acc. then it will be $(g–a)\sin\theta$.
But in the problem acceleration = –a(retardation)
 Acceleration of block = $[g – (–a)\sin\theta = (g + a)\sin\theta]$.

16) Frictional force is independent of the area of surfaces in contact, as long as normal reaction between the two bodies in contact remains the same.
So, $f_1 = f_2 = f$

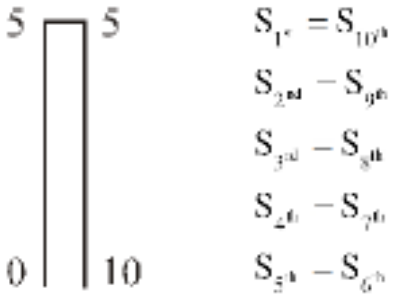
17) Force on the football $F = m\frac{dv}{dt}$
$$F = \frac{m(v_2 - v_1)}{dt} = \frac{0.5 \times (10 - 0)}{1/50} = 250\ N$$

18) $\vec{F} = \frac{d\vec{p}}{dt}$



19) 130 T = 130 N

20) $M^0 - \frac{\Delta m}{\Delta t} \cdot t$
Hence option (1)



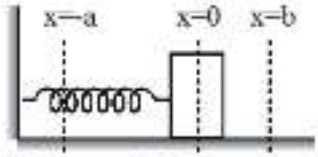
21) $S_{1^{st}} = S_{10^{th}}$
$S_{2^{nd}} - S_{9^{th}}$
$S_{3^{rd}} - S_{8^{th}}$
$S_{4^{th}} - S_{7^{th}}$
$S_{5^{th}} - S_{6^{th}}$
so Answer will be Option (1)

22) $\frac{g\sin\theta}{g\sin\theta - \mu g\cos\theta} = \frac{2}{1}$, $\theta = 45°$
$\frac{1}{1-\mu} = \frac{2}{1} \Rightarrow \mu = 0.5$
Hence option (1)

23) $F_L = \mu_S mg$
$200 = \mu_s \times 500$
$\mu_s = 0.4$
$f_k = \mu_k mg$
$150 = \mu_k \times 500$
$\mu_k = 0.3$

24) Only 2kg will move.
for 2 kg a = $\frac{10-4}{2} = 3m/s^2$

25) $T_1 = m_A g$ and $T_2 = \frac{2 \times m \times 3m}{4m} g$
but $T_1 = 2T_2$ so $m_A = 3m$



$W_{total} = \Delta KE$

$$\int_{-a}^{b} -Kx\,dx + \mu mg(a + b)\cos 180^\circ = 0 - 0$$

$$-K\left[\frac{b^2}{2} - \frac{a^2}{2}\right] - \mu mg(a + b) = 0$$

$$\frac{K}{2}(a - b)(a + b) = \mu mg(a + b)$$

26) $(a - b) = \dfrac{2\mu mg}{K}$

27) $R = \sqrt{(mg\cos\theta)^2 + (mg\sin\theta)^2} = mg$

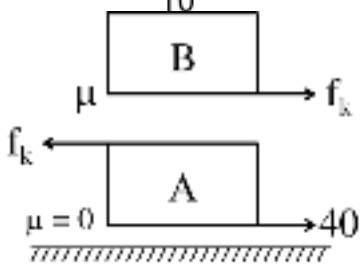
28) $a_{common} = \dfrac{40}{15} = \dfrac{8}{3} m/s^2$

$(a_B)_{max} = \mu g = 2 m/s^2$

As $(a_B)_{max} < a_{common}$

There will be slipping

$a_A = \dfrac{40 - 0.2 \times 5 \times 10}{10} = 3 m/s^2$



29)

Let speed of bus = v_B,

speed of scooty = v_S = 60 km/h

and distance between two consecutive buses = d Now according to question

$d = (v_B + v_S)(10 \text{ min}) = (v_B - v_S)(30 \text{ min})$

$\Rightarrow v_B + v_S = 3v_B - 3v_S \Rightarrow v_B = 2v_S = 120$ km/h

Time period, $T = \dfrac{d}{v_B} = \dfrac{(v_B + v_S)(10 \text{ min})}{v_B}$

$= \dfrac{3}{2}(10 \text{ min}) = 15 \text{ min}$

30) K.E. acquired by the body = work done on the body K.E. = Fs

i.e. it does not depends upon the mass of the body.

31)

From (0, 0) to a, 0

$y = 0 \Rightarrow W_1 = \int_0^a \vec{F}.\vec{dx}$

$W_1 = -\int_0^a Kxdx = \frac{-Ka^2}{2}$

From (a, 0) to (a, a)

$W_2 = \int_0^a -K\left(a\hat{i} + y\hat{j}\right)\left(dy\hat{j}\right)$

$W_2 = -\left(\frac{Ky^2}{2}\right)_0^a$

$W_2 = -\frac{Ka^2}{2}$

$W = W_1 + W_2 = -Ka^2$

32) $W_f = (-\mu mg \cos\theta)d$

$= -0.20 \times 5 \times 10 \times \frac{1}{\sqrt{2}} \times 10$

$= \frac{-2 \times 50}{\sqrt{2}} = -50\sqrt{2}$

33)

Work output of engine = mgh

$= 100 \times 10 \times 10 = 10^4$ J

Efficiency (η) = $\frac{\text{output}}{\text{input}}$

 Input energy = $\frac{\text{output}}{\eta}$

$= \frac{10^4}{60} \times 100 = \frac{10^5}{6}$ J

 Power = $\frac{\text{input energy}}{\text{time}} = \frac{10^5/6}{5} = \frac{10^5}{30}$

= 3.3 kW

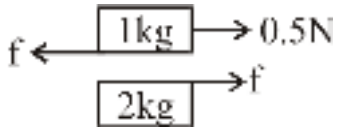
34) $S = \frac{t^3}{3} \Rightarrow v = t^2$

$v_1 = 0$, $v_2 = (2)^2 = 4$ m/s

$W = \Delta K = \frac{1}{2}(3)(4)^2 = 24$ J

35)

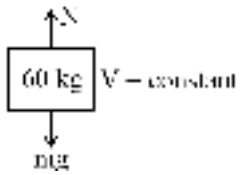
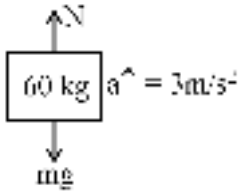
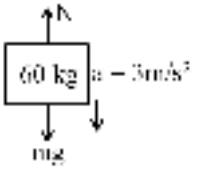
mg – T = ma
mg = T
T = 2000
W_T = –T.S
= –2000 × 20 × 2
= –80000 = $–8 \times 10^4$ J



36)
0.5 = 3a ⇒ a = 5/30
f = 2a = 2(5/30)= 1/3N
f_{max} = 0.1 × 10 = 1N
 f = 1/3 N will act
work done by lower block
= –f(s)
= –1/3(3) = –1 Joule

37) $F_{ext.}$ = = 0, $W_{ext.} = \vec{F}_{ext}.\vec{s} = 0$
By work-energy theorem,
W = K_F – K_I
$W_{ext.} + W_{int.} = 0 - \left(\frac{1}{2}mv^2 + \frac{1}{2}mv^2\right)$
$W_{int.}$ = – mv^2 = – 5 × 2^2 = – 20 J.

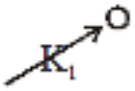
38)

 N = mg N = 600 N	 N – mg = ma N = m(g + a) N = 60(10 + 3) N = 780 N	 N = m (g – a) N = 60 (10 – 3) N = 60 × 7 N = 420 N

39) v = u + at
$v_0 = 0 + at_0$
$a = \frac{v_0}{t_0}$
$v = u + at = 0 + \frac{v_0}{t_0} t$
$\Rightarrow v = \frac{v_0}{t_0}t$

$$w = \Delta kE = \frac{1}{2}, m\frac{v_0^2}{t_0^2}, t^2$$

40) Mechanical energy conservation between A & P

$U_1 +$  $= K_2 + U_2$

$mg \times 2 = mg \times 1 + K_2$

$K_2 = mg \times 1 = 10$ J.

41) $v = k\sqrt{s}$

$$a = k\left(\frac{1}{2\sqrt{s}}\right) \times k\sqrt{s}$$

$$\boxed{a = \frac{k^2}{2}}$$

$$s = 0 + \frac{1}{2}\left(\frac{k^2}{2}\right)t^2$$

$$W = \frac{mk^2}{2} \times \frac{k^2t^2}{4}$$

42) In first case acceleration of block is zero always.

Therefore $mg = kx$

$\therefore \quad x = \frac{mg}{k} = d$(given)

In second case block variably accelerate from rest to rest

So 0 (zero) $= -mgx + \frac{1}{2}kx^2$

$\therefore \quad x = \frac{2mg}{k} = 2d$

43) $[\beta] = T$

$$ML^2T^{-3} = \frac{\alpha v}{\beta} \Rightarrow \left(\frac{\alpha}{\beta}\right) = \frac{ML^2T^{-3}}{LT^{-1}}$$

$$\left(\frac{\alpha}{\beta}\right) = MLT^{-2}$$

44)

$$[D] = [n]\frac{[x_2 - x_1]}{[n_2 - n_1]} = \left(\frac{1}{L^2T}\right)\left(\frac{L}{\frac{1}{L^3}}\right) = [L^2T^{-1}]$$

45) 10 MSD = 11 VSD

1 VSD $= \frac{10}{11}$ MSD

LC = 1MSD – 1VSD

$= 1 \text{ MSD} - \frac{10}{11} \text{ MSD}$

$= \frac{1\text{MSD}}{11}$

$= \frac{5}{11}$ units

CHEMISTRY

46) $\Delta S_{vap} = \frac{\Delta H_{vap}}{T_b}$

Using $T_b = 373$ K

$\Delta S_{vap} = \frac{186.5 \text{kJmol}^{-1}}{373\,\text{K}} = 0.5\,\text{kJ K}^{-1}\,\text{mol}^{-1}$

(Using $T_b = 373.15$ K, $\Delta S_{vap} = \frac{186.5\,\text{kJ mol}^{-1}}{373.15\,\text{K}}$

» 0.500067 kJ K^{-1} mol^{-1}, which round to 0.5 kJ K^{-1} mol^{-1})

47)

$\Delta H = \Sigma\Delta H_{f(P)} - \Sigma\Delta H_{f(R)}$

or $\Delta H = [81 + 3(-393)] - [9.7 + 3(-110)]$

or $\Delta H = -1098 + 320.3$

or $\Delta H \simeq -778$ kJ mol^{-1}

48)

$CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$

$\Delta H = (\Delta H)_P - (\Delta H)_R$

$\Delta H = -178$ kJ/mol

$\Delta H = \Delta E + \Delta ngRT$

49)

Calorific value = $\frac{\Delta H_{comb}}{\text{Molar Mass}}$

50) $P(s) + \frac{3}{2}Cl_2(g) \rightarrow PCl_3(g)$

$306 = 314 + \frac{3}{2} \times 2 \times 121 - 3E_{P-Cl}$

$3E_{P-Cl} = 371$

$E_{P-Cl} = 123.66$ kJ/mol

51)

w = –2.303 nRTlog$\left(\frac{V_2}{V_1}\right)$

52) ΔH – TΔS = 0

53) $\Delta H = \Delta E + \Delta ng\ RT$

54) **Question Explanation :**
Calculate the free energy change (ΔG) for a reaction at 1000°C, given ΔH and ΔS.

Given Data :
ΔH – 31400 cal
ΔS = 32 cal K^{-1} mol^{-1}
Temperature (T) = 1000°C

Concept :
Gibbs Free Energy Change (ΔG = ΔH – TΔS).

Mathematical Calculation :
1. Convert Temperature to Kelvin (K):
T = 1000°C + 273.15 = 1273.15 K. (Using 1273 K for simplicity as options are integers).
T = 1273 K.

2. Calculate ΔG:
ΔG = ΔH – TΔS
ΔG = 31400 cal – (1273 K × 32 cal K^{-1} mol^{-1})
ΔG = 31400 cal – 40736 cal mol^{-1}
(Assuming the given ΔH is per mole for consistency with ΔS)
ΔG = – 9336 cal mol^{-1}

Final Answer :
The free energy change (ΔG) is –9336 cal. Option (1)

55)

Explaining Question:
Find volumes of CH_4 and C_3H_8 in a 6 L mixture producing 7 L CO_2 on combustion.
Concept: This question is based on Volume ratios from balanced combustion equations (Gay-Lussac's law)
Solution:
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \rightarrow$ 1 L CH_4 gives 1 L CO_2
$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O \rightarrow$ 1 L C_3H_8 gives 3 L CO_2
Let CH_4 = x L, then C_3H_8 = 6 − x L
Total CO_2 = x + 3(6 − x) = 7
⇒ x + 18 − 3x = 7
⇒ −2x = −11 → x = 5.5
CH_4 = 5.5 L, C_3H_8 = 0.5 L

56) **Question Explanation :**
Identify conditions for a chemical reaction to be impossible (non-spontaneous) at any temperature.

Concept :
Gibbs Free Energy ($\Delta G = \Delta H - T\Delta S$) and Non-Spontaneity ($\Delta G > 0$).

Reasoning :
A reaction is non-spontaneous if $\Delta G > 0$.
Consider $\Delta G = \Delta H - T\Delta S$. For ΔG to be always positive (for $T > 0$):
If ΔH is positive (unfavorable) and ΔS is negative (unfavorable):
ΔG (positive value) – T(negative value)
ΔG (positive value) + (positive value from $T \times \Delta S_{abs}$)
Thus, ΔG will always be positive.

Final Answer :
A spontaneous reaction is impossible at any temperature when ΔH = +ive, ΔS = –ive. Option (2)

57) $CH_3COOH \rightleftharpoons CHCOO^- + H^+$

1–x	x	x + 0.1
≃1		≈0.1

$$2 \times 10^{-5} = \frac{x \times 0.1}{1}$$

$x = [CH_3COO^-] = 2 \times 10^{-4}$ M

58) **Given:**
- Reaction is exothermic
- Gaseous reactants and products
- Changes include temperature, pressure, catalyst, inert gas, and removal of product

Concept:
Le Chatelier's Principle
Explanation :
(a) Increase of temperature: Shifts equilibrium backward (exothermic) decreases yield
(b) Increase of pressure. Total gas moles decrease (2 →1), so equilibrium Shilts forward increases yield
(c) Addition of catalyst: No effect on equilibrium, only increases rate no change in yield
(d) Addition of inert gas at constant volume: No effect on partial pressures ⇒ no change in yield
(e) Removal of C. Equilibrium shifts forward to replace C ⇒ increases yield
Final Answer: 3. b & e
Only increase in pressure and removal of product C increase the yield.

59)

$N_2 + O_2 \rightleftharpoons 2NO$

1	1	0
1–∝	1–∝	2∝

$K_c = \frac{(2\alpha)^2}{(1-\alpha)^2}$ or $\sqrt{K} = \frac{2\alpha}{1-\alpha}$

or $\sqrt{K} - \sqrt{K}\alpha = 2\alpha$ or $\sqrt{K} = \alpha\left(2 + \sqrt{K}\right)$

$\alpha = \frac{\sqrt{K}}{2+\sqrt{K}}$

60)

$$Q_p = \frac{(P_{NO})^2}{\left(P_{N_2}\right) \times \left(P_{O_2}\right)} = \frac{\left(4 \times 10^{-3}\right)^2}{0.5 \times 0.2}$$

$= 1.6 \times 10^{-4}$ $Q_P < K_P$

Reaction will move forward

61)

Explain the Question :

We need to check if pH of water increases with temperature and whether the reason (endothermic dissociation of water) explains it.

Concept : This question is based on Ionization of water ($H_2O \rightleftharpoons H^+ + OH^-$) is endothermic.

Solution : With increase in T, Kw increases → $[H^+]$ increases → pH = –log$[H^+]$ decreases (not increases).

So, assertion is false but reason is true.

Final Answer : 3

62) $2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$

At equ. P – $\frac{3P}{9}$ $\frac{6P}{9}$ $\frac{2P}{9}$ $\frac{P}{9}$

$P_T = \frac{6P}{9} + \frac{2P}{9} + \frac{P}{9} = P$

$K_p = \frac{\left(\frac{2P}{9}\right)^2 \times \left(\frac{P}{9}\right)}{\left(\frac{6P}{9}\right)^2} = \frac{K_P}{P} = \frac{1}{81}$

63)

(1) + (2) = (3)

$K_1.K_2 = K_3$

64)

$K_a = C\alpha^2 \Rightarrow \alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5} \times 2}{0.1}}$

$= 1.41 \times 10^{-2}$

Percentage dissociation = α × 100%
= 0.0141 × 100% ≈ 1.41%

65) **Relevant Formulas:** At a fraction f of neutralization,

$$pH = pK_a + \log\left(\frac{f}{1-f}\right)$$

where - pH = measure of acidity/basicity - pKâ,
= negative log of the acid dissociation constant - f = fraction of acid neutralized

Step-by-Step Calculation: For the fraction of neutralization $f_1 = \frac{1}{5}$,

$$pH_1 = pK_a + \log\left(\frac{1/5}{1-1/5}\right)$$

$$= pK_a + \log\left(\frac{1/5}{4/5}\right) = pK_a + \log\left(\frac{1}{4}\right)$$

For the fraction $f_2 = \frac{4}{5}$,

$$pH_2 = pK_a + \log\left(\frac{4/5}{1-4/5}\right)$$

$$= pK_a + \log\left(\frac{4/5}{1/5}\right) = pK_a + \log 4$$

The difference in pH is :
$\Delta pH = pH_2 - pH_1 = (pK_a + \log 4)$

$- (pK_a + \log\frac{1}{4}) = \log 4 - (-\log 4) = 2 \log 4$

66)

Explaining the Question:
pH of resulting solution
Concept: Buffer solution
Solution: (a) Given data → Ka = 1.8 × 10–4
KOH → .2M, 50 ml → 10 m mol
HCOOH → .5M, 40 ml → 20 m mol
(b) Rx^n →

HCOOH + KOH →	H COOK	+ H_2O	
20	10	0	–
10	0	10	–

pH of Acidic buffer

$$= p^{ka} + \log\frac{S}{A}$$

$$= 4\text{-}\log 1.8 + \log\frac{10}{10}$$

= 3.75 **Final Answer:** Option_1

67)

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68) $pH = pK_a + \log \frac{[CB]}{[WA]}$
if $pH = 4.74 \Rightarrow pH = pK_a$
acid is half neutralised
$0.2 \times v = 0.2 \times \frac{50}{2}$
$v = 25$ mL

69) **Solution/Explanation/Calculation:**
Concept:
$Ca_3(PO_4)_2 \rightleftharpoons 3Ca^{+2} + 2(PO_4)^{-3}$
Final Answer: $K_{sp} = [Ca^{+2}]^3 \ [PO_4^{-2}]^2$

70) Degree of hydrolysis of salt of weak acid and weak base is independent from concentration.

71) $K_{sp} = (S)\ (0.2)^2$
$\Rightarrow S = 0.45 \times 10^{-9}$

72) After mixing
$[Ag^+] = 10^{-4} \times \frac{V}{3V} = 3.3 \times 10^{-5}$
$[Cl^- -] = 10^{-6} \times \frac{2V}{3V} = 6.6 \times 10^{-7}$
 I.P. $= [Ag^+][Cl^-]$
$= 3.3 \times 10^{-5} \times 6.6 \times 10^{-7}$
$= 2.2 \times 10^{-11}$
 I.P. $< K_{sp}(10^{-6})$ Precipitation will not occur.

73)

$\Delta G_{system} = -T\Delta S_{total}$ is correct relation

74) **Question Explanation:**
We need to find ΔH of formation of carbon monoxide.
Concept:
This question is based on Enthalpy of formation.
Solution:
$C(s) + O_2(g) \rightarrow CO_2(g)$; $\Delta H = -393.5$ KJ
Combustion of carbone monoxide
$CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$ $\Delta H = -283$ KJ
Reverse the second reaction and add:
$CO_2(g) \rightarrow CO(g) + \frac{1}{2}CO_2(g) \Delta H = 283KJ$
$C(s) + O_2(g) \rightarrow CO_2(g) \Delta H = -393.5KJ$

$$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) \Delta H = 283 - 393.5 = -110.5 KJ$$

Correct Option: 2

75)

Explaining the Question : We need to find correct relation between assertion and reason.
Concept : Spontaneous process and ΔS

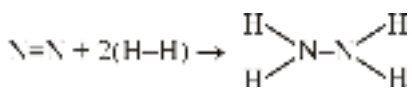
Solution : Assertion :- true because spontaneous process proceed in one direction naturally and require external inter vention to reverse.
Reason :- True (ΔS total of a spontaneous process is positive)
Assertion and reason are true both and reason is not the correct explanation of assertion

Final answer : Option (2)

76) NCERT-XI, Pg. # 183, Q. no. 6.19

77) $\Delta H = \Delta E + \Delta n_g RT$
$40 = \Delta E + (2 \times 2 \times 10^{-3} \times 500)$

78) **Asking About :**
ΔE/internal energy change for process
Concept :
Internal energy change is defined for 1 mole.
Solution/Explanation :
Given :
→ CH_4 (mass) = 0.4 gm → Heat evolved (q_v) = – 100 KJ (since bomb Calorimetry operates at constant volume $q_v = \Delta E$)
→ Molar mass of CH_4 = 16 g/mol
Acc. to question 0.4 gm CH_4 on
combustion produces → 100 KJ
To determine ΔE per mole
For 1 mole (i.e. 16 gm methane)
0.4 → – 100 KJ
16 gm → $= \frac{100}{0.4} \times 16$ ΔE = – 4000 KJ/mole
Option (2) is correct answer.

79) 

$\Delta_f H^o_{N_2H_4} = \Delta_{rxn} H^o$
= Σ B.E. of reactant – Σ B.E. of product
= (941 + 2 × 436) – (159 + 4 × 398)
= 62 kJ/mol

80)

$4C(s) + 5H_2(g) \rightarrow C_4H_{10}(g)$

$\Delta H_{reaction} = (\Sigma \Delta H_{comb})_R - (\Sigma \Delta H_{comb})_P$

$(\Delta H_{comb})_{C_{(s)}} = (\Delta H_f)_{CO_2}$

$(\Delta H_{comb})_{H_2} = (\Delta H_f)_{H_2O}$

$= [4 \times (-393.5) + 5 \times (-285.8)] - [-2878]$

$= -125$ kJ/mol

81) $CO_{2(g)} + C \rightarrow 2CO_{(g)}$

1L	–	-
1 – x		2x

Total $\Rightarrow 1 - x + 2x = 1.5L \Rightarrow (x = 0.5L)$

$V_{CO} = 2x = 1L$ & $n_{CO} = \frac{1}{22.4}$ mole at STP

82) **Question Explanation:**
We are asked to calculate the percentage purity of a $CaCO_3$ sample that produced 3.3 g of CO_2 when treated with excess HCl.
Concept:
Stoichiometry – Law of chemical combination
Solution:
Reaction: $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$
Molar mass: $CaCO_3 = 100$ g, $CO_2 = 44$ g
1 mole $CO_2 \rightarrow 1$ mole $CaCO_3$
44 g $CO_2 \rightarrow 100$ g $CaCO_3$
3.3 g $CO_2 \rightarrow \frac{100}{44} \times 3.3 = \frac{30}{4} = 7.5g$ $CaCO_3$
% purity of $CaCO_3 = \frac{7.5}{10} \times 100 = 75\%$
Final Answer : option (4)

83) $\underset{(+7)}{MnO_4^-} + \underset{(+2)}{Fe}\ \underset{(+3)}{C_2O_4} \rightarrow \underset{(+2)}{Mn^{+2}} + \underset{(+4)}{CO_2} + \underset{(+3)}{Fe^{+3}}$

eq. of oxidant = eq. of reductant

$n \times 5 = 10 \times (1 \times 1 + 2 \times 1)$

$\therefore$ moles of $MnO_4^- = \frac{30}{5} = 6$

84)

$2BrO_3^- + 12H^+ + 10e^- \rightarrow Br_2 + 6H_2O$

$2Br^- \rightarrow Br_2 + 2e^-$

$2BrO_3^- + 10Br^- + 12H^+ \rightarrow 6Br_2 + 6H_2O$

Eq wt. $= \frac{6M}{10} \Rightarrow \frac{3M}{5}$

85) NCERT XI, Part-II, Pg # 244
Revised edition, October 2022

86)

Theoretically learn Question.

87)

For $H_4P_2O_5$
$\Rightarrow +4 + 2x - 10 = 0$
$x = +3$
For $H_4P_2O_6$
$+4 + 2x - 12 = 0$
$x = +4$
For $H_4P_2O_7$
$+4 + 2x - 14 = 0$
$x = +5$

88) $$Q = \frac{[SO_2Cl_2]}{[SO_2][Cl_2]} = \frac{\frac{0.8}{2}}{\left(\frac{0.4}{2}\right)\left(\frac{0.2}{2}\right)} = 20$$
$\because$ $K_c = 10$ therefore , $Q > K_c$
and backward reaction will proceed to attain equilibrium.

89)

	NH_4OH	+ HCl	$\rightarrow$ NH_4Cl	+ H_2O
t = 0	5	4	0	0
t = eq	1	0	4	4

Mixture of (NH_4OH + NH_4Cl) is basic buffer solution.

90) • The choice of an indicator depends on the type of acid-base reaction.
• Strong acid + strong base titrations have an equivalence point at pH ≈ 7.
• Phenolphthalein (pH range: 8.2 – 10) is suitable for strong base-weak acid titrations.
• Methyl orange (pH range: 3.1 – 4.4) is ideal for strong acid – weak base titrations.
• For strong acid-strong base (NaOH + HCI), any indicator (pH 3 – 10) can work.
Final Answer: Option (4) All

BIOLOGY

91) NCERT Pg. # 46

92)

NCERT-XI, Pg. # 44

93) NCERT XI Pg # 29, 31, 32
a, c, d are haploid in nature.

94) NCERT XI Pg No. # 82,83 (Fig. 7.3, 7.4)

95)

NCERT Pg # 19

96) NCERT XI ; Page No. # 28

97)

NCERT, Pg. # 50

98)

Only male frog

99) NCERT XI ; Page No. # 17,18

100) NCERT Pg.#26-33.

101) NCERT XI ; Page No. # 14

102) NCERT Pg.#103.

103) NCERT-XII, Pg. # 104

104) NCERT XI, Pg. # 49, 4.2.1

105) NCERT XI ; Page No. # 12,13,14

106) NCERT-XI, Pg. # 14

107) NCERT XI ; Page No. # 32

108)

A pair of yellowish ovoid testes

109) NCERT-XI, Pg. # 13

110)

Question Explanation: Identifying forms and life cycle of Obelia.
Concept: Alternation of generation in Obelia.
Solution: The two basic body forms of **coelenterates** are the **polyp** (sessile, cylindrical form, represented by A) and the **medusa** (free-swimming, umbrella-shaped form, represented by B). In Obelia, the polyp reproduces **asexually** to produce the medusa, and the medusa reproduces **sexually** to form the polyp, a phenomenon known as alternation of generation.
Final Answer: (A) produce (B) asexually and (B) produce (A) sexually

111)

NCERT XI Pg # 49, 50, 55, 59

112)

Ventral side of skin is generally olive green

113) NCERT XI ; Page No. # 17

114) NCERT Pg.# 82

115) NCERT Pg.#112 (E), 112 (H)

116) NCERT XI ; Page No. # 30,31,32

117)

NCERT Page No. 82

118) NCERT Pg. # 28

119) NCERT Pg # 17,18

120) NCERT, Pg. # 5, 23, 24

121) NCERT XI Pg. # 16,17,18

122) NCERT Pg. # 58

123) NCERT-XII, Pg. # 102

124)

NCERT Pg. # 12, 13

125) NCERT XI ; Page No. # 30

126) NCERT XI Pg # 32

127) NCERT-XII, Pg. # 54

128)

NCERT-XI, Pg. No. # 29, 32,33, 34

129) NCERT XI Pg.# 13

130)

All of these

131) NCERT Page # 16

132) NCERT XI ; Page No. # 29

133) NCERT Pg. # 119

134) NCERT XI ; Page No. # 91

135) NCERT Pg. # 29,30

136) NCERT XI, Pg. # 101, Last para

137) NCERT XI (E), Page No. 104

138)

NCERT Pg. No. # 54

139) NCERT XI Page No. 32

140) NCERT Pg. # 113

141) NCERT Page No. # 52-54

142) NCERT-XI - Page No.-49

143) NCERT Page # 17

144) NCERT, Pg. # 32

145)

Both nucleus and haemoglobin

146) NCERT Pg # 54

147) NCERT Pg. # 59, 60

148) NCERT Pg. # 83

149) **Question Explanation:** Identifying exclusively marine phyla.
Concept: Animal habitats.
Solution: Ctenophora and Echinodermata are phyla with exclusively marine members. Mollusca, Annelida, and Chordata also include freshwater and terrestrial species.
Final Answer: 4

150)

NCERT-XI, Pg # 30

151)

NCERT-XI, Pg. No. # 29, 32, 33
A, B, C & D are correct.

152) NCERT Pg.3112 (E), 113 (H)

153) NCERT XI ; Page No. # 20,21

154) NCERT Pg. No. # 114

155) NCERT-XI Pg. # 112

156) NCERT XI Page No. 18

157) NCERT Pg.#39-45.

158)

NCERT XI Pg. # 7

159) NCERT Pg. # 15

160) NCERT Pg. # 26

161) NCERT-XI, Pg#115, Ist PARA

162) NCERT XI ; Page No. # 15

163) NCERT Pg. # 38, 39

164) *Aspergillus*, Yeast, *Claviceps*, morels are
Ascomycetes

165) NCERT-XII, Pg. # 57

166) NCERT Pg.#42-45

167)

NCERT Pg.#26-34
Salvinia, Selaginella, Cycas, Mango, *Pinus*

168) NCERT-XI, Pg. # 20

169)

NCERT XI, Pg # 32

170) NCERT Pg.#XI- 40,41,42,43.

171) NCERT XI Page No.: 26, 30

172) NCERT Pg.#15

173)

NCERT-XI, Pg # 26, 27 & 29

174) NCERT XI ; Page No. # 32,33

175)

The correct answer is 4. To move particles and mucus in a specific direction over epithelium.

Here's why : Cilia are hair-like projections on the surface of some epithelial cells. They beat in a coordinated manner to move fluids, particles, or mucus across the cell surface.

176) NEERT Page No. # 57

177) **Explaining :** (i) Testes, (ii) Adrenal gland, (iii) Fat bodies, (iv) Cloaca gland.
Concept :
Final Answer : (1)

178) NCERT Pg.#114 (E), 114 (H)

179) NCERT, Pg # 52,43,54

180) NCERT XI, Pg # 26